Chapter 1: Background and Methodological Framework

1 Data Description

Before describing the dataset used to evaluate student performance, it is necessary to provide an overview of the research context. Throughout this study, three major indicators were used to evaluate students' violin-learning progress: Progress Index, Total Number of Issues, and Practice Duration. The breakdown of these three major indicators is as follows:

1.1 Progress Index

The main textbook used in this study is the *Shinozaki Violin Method*. Students start learning with the *Shinozaki Violin Method for Children* and, upon completion, transition directly to page 23 of *Shinozaki Violin Method Volume 1* to continue learning (since the content before page 23 overlaps with the children's edition). Below are the learning contents corresponding to each progress index value:

Table 1: Progress Index and Corresponding Learning Content

Progress Index	**Learning Content**
3	Introduction to the violin, holding the bow, attaching the shoulder rest, holding the violin with the chin/shoulder, learning the letter names and solfège of the four strings, the musical staff, and basic preparations before taking out the instrument to play.
4–7	1. Learning quarter notes, half notes, and their corresponding rests. 2. Bowing exercises on the E and A strings.
8	1. Learning and practicing the 1st finger on the A string (B). 2. Learning dotted half notes.
9–11	Learning and practicing the 2nd finger on the A string (C♯), alongside combined exercises featuring the open A string, 1st finger, and 2nd finger.

Progress Index	Learning Content
12–15	Learning and practicing the 3rd finger on the A string (D), alongside combined exercises for the open A string, 1st finger, 2nd finger, and 3rd finger.
16–18	Combined exercises for the open A and E strings and fingers 1–3 on the A string.
19–24	Learning and practicing the 1st finger on the E string (F♯), alongside mixed exercises with previously learned fingers.
25	Learning and practicing the 2nd finger on the E string (G♯), alongside combined exercises for the open E string, 1st finger, and 2nd finger.
26–27	Learning and practicing the 3rd finger on the E string (A), along with combined exercises for the open A/E strings and fingers 1–3.
28–29	Eighth note exercises.
30	Dotted quarter note exercises.
31–33	1. Combined exercises for the open A string and fingers 1–3, including low 2nd finger on the A string (C natural). 2. Learning bowing techniques (whole bow, upper/middle/lower half bow) and practicing with a metronome.
34–35	1. Understanding that the 4th finger on the A string equals the open E string (E). 2. Exercises combining the open A string and all fingers.
36–38	Learning and practicing the open D string, notes for each finger on the D string, and exercises on the D string.
39–43	Mixed finger exercises on the A and D strings.
44–45	Detaché and slur exercises.
46–47	Learning and practicing the open G string, notes for each finger on the G string, and exercises on the G string.
48–52	Mixed finger exercises across the A, D, and G strings.
53–55	Mixed finger exercises across the A, D, and G strings, incorporating a large volume of eighth notes.
56–57	Finger exercises on the E string and learning finger positions for F natural and G natural.
58–60	All-string mixed exercises (C Major).
61	Staccato exercises.
62	Basic exercises and etudes in A minor.
63–67	Basic exercises and etudes in G Major.
68	Basic exercises and etudes in E minor.

Progress Index	Learning Content
69–72	1. Basic exercises and etudes in D Major. 2. Basic dotted eighth note exercises (72).
73	Basic exercises and etudes in B minor.
74–75	Sixteenth notes and related rhythmic pattern exercises.
76–77	Basic exercises and etudes in F Major.
78	Basic exercises and etudes in D minor.
79	Basic exercises and etudes in B♭ Major.
80	Basic exercises and etudes in G minor.
81–82	”Young Wings” (The final piece before entering *Shinozaki Volume 2*).

1.2 Number of Issues

During lessons, this study listed several common mistakes regarding students' learning issues. The most common problems observed in students include:

- **Neck**: Left wrist collapsed against the neck of the violin.
- **Bow Hold**: Posture slips or becomes non-standard during bowing, or even gripping/clutching the bow while playing.
- **Elbow**:
  - **Right Elbow**: The right elbow should adjust to corresponding heights when playing on different strings. A common issue is a student failing to master the proper elbow height for a specific string, resulting in scratchy or unwanted noise.
  - **Left Elbow**: When left-hand fingers press the strings, the elbow should aim toward the right side. If it stays too far to the left, fingers will not reach high positions; young children may even struggle to press notes on the G string in 1st position.
- **Wrist**: When bowing downward (down-bow), the right arm should extend; when bowing upward (up-bow), the right wrist needs to bend. Failing to bend the wrist affects tone quality and causes the elbow to assume an unnatural height, easily generating scratchy sounds.
- **Bowing Direction**: When bowing, the bow should stay perpendicular to the strings and move midway between the bridge and fingerboard, a feat that is quite difficult for beginners to achieve.

- **Sight-Reading**: The correct sequence for sight-reading is: Eyes see the note $\rightarrow$ Read/vocalize the note $\rightarrow$ Play the note. Although this process occurs instantaneously, following this specific order improves sight-reading skills. When starting a new piece, playing smoothly is usually hindered by skipping sight-reading or following an incorrect sequence.
- **Intonation**: The ability to produce accurate pitch; errors occur when notes are played out of tune.
- **Violin Hold**:
  - Rotating the instrument during bowing to force a perpendicular alignment between bow and string. This only makes the bow look perpendicular on the surface. In reality, the player's body rotates with the instrument, making bowing unstable.
  - Clamping down with the chin while holding the violin (the correct method is setting the violin on the shoulder first), causing the instrument to droop.
- **Rhythm/Timing**: Understanding the theoretical duration of notes, but executing the note durations incorrectly while playing.
- **Tilted Bow**: The bow hair should stand upright during bowing. For beginners, if the bow tilts outward, it is not a tone technique choice, but rather a result of unstable bowing or poor sound production.
- **Tempo**: Having the ability to comprehend and execute rhythm, but struggling with tempo control (speeding up, slowing down, or playing at erratic speeds).
- **Bowing Pressure**: The right hand must hold the bow securely and apply balanced perpendicular pressure. Too little force causes the bow to slide; too much force (usually due to gripping the bow too tightly) causes the sound to crack.
- **Finger Strength**:
  - Young beginners may lack finger strength in the left hand to press the strings firmly, resulting in unwanted noise.
  - Fingers failing to stand curved/upright on the strings; even if placed on the correct spot, the produced pitch remains out of tune.
- **Fingering**: Mainly an unfamiliarity with the sequence: Reading the note $\rightarrow$ Playing the note.

- **Focus/Attention**: As the name suggests, a lack of concentration during lessons.
- **Bowing Technique**: Includes up/down bow directives, whole/half bow placements, and detaché/slur markings. Executing a stroke that differs from the assigned bowing causes significant confusion in group settings or orchestras.
- **Swaying**: Swaying the body while playing. This primarily stems from a lack of arm strength in young children, who try to use their entire body to assist in bowing.
- **Other**: Issues not covered in the above categories or unexpected temporary situations occurring during class.

1.3 Practice Duration

We denote practice time with 0.5 hours per unit.

2 Data Collection Procedure

Each student was evaluated once a week across the three primary criteria:

1. **Course Progress**: If a student was an absolute beginner who had not yet started textbook material, the value was recorded as 3. For the *Shinozaki Violin Method for Children*, the value matched the page number. For *Shinozaki Volume 1*, the value was calculated as Page Number + 8 (e.g., the final piece, "Young Wings," was recorded as 81 and 82). Values for *Shinozaki Volume 2* began at 83 onwards. However, because the data in this study did not reach Volume 2, methods for tracking Volume 2 are omitted.

   A student's performance was held to the benchmark of *"playing smoothly at a fixed tempo using the designated technique."* Only after reaching this standard could they advance to the next numerical progress step.

2. **Common Learning Problems**: Lesson conditions were monitored for the presence of the 18 common problems listed above, categorized into 3 severity levels:

   - **1.0**: Assigned if one of the 18 problems appeared and required multiple reminders or extended time to correct during class, indicating that the issue was clearly observed.

- **0.5**: Assigned if the problem appeared but was corrected after one or two reminders without recurring, or if performance improved compared to the previous week, indicating that the issue might be present.
- **0** (or left blank): Assigned if the problem did not occur, indicating that the issue was not observed.

After evaluating each item individually, the total number of problems present in that lesson was summed, ranging within $[0, 18]$ with increments of 0.5.

3. **Practice Time**: Recorded practice time between classes with 0.5 hours per unit.

In addition to these three core elements, the student's ID number, age, lesson count, and date were required. Table 2 presents an example of a single observation in the dataset.

Table 2: Example of a Single Observation in Dataset

Date	Lesson #	ID	Age	Neck	Bow Hold	Elbow	Wrist	Bowing Dir.	Sight-Read	Intonation	Violin Hold	Rhythm	Tilted Bow	Tempo	Pressure	Finger Str.	Fingering	Focus	Bowing Tech.	Swaying	Other	Progress	Practice Time	Total Issues
2017.9.1	1	1	6		1			1	1													4	1	3

3 Study Population and Teaching Context

In this study, students were categorized into two groups: those who began violin training at age 6 and those who began training at age 7. Age 6 beginners means the student starts to learn the violin at age 6, and Age 7 beginners means the student starts to learn the violin at age 7. However, all students in the age-7 beginner group had received limited violin exposure at age 6. In this case, we did not change our beginning textbook page, but we estimated the extent of their prior knowledge based on the available data.

We have 14 students in this report. Due to privacy considerations, we do not show students' names and personal information, including their specific reasons for withdrawal. Table 3 provides the available background information.

During the learning process, students occasionally encountered performance-related challenges that required additional preparation over several lessons. These preparation periods temporarily affected their textbook progression. Therefore, this study examined the relationship between textbook progression and students' underlying performance levels during performance preparation periods. Except for several necessary pieces of contextual information, for example, Students 3–6 participated in performance preparation from September to November 2017, which influenced the estimation of their initial prior knowledge, all other

Table 3: Student Demographics and Status Overview

Student ID	Age at Entry	Enrollment Time	Current Status	Withdrawal Time
1	6	Sep 2017	Active	
2	6	Sep 2017	Dropout	Aug 2018
3	7	Jul 2017*	Active	
4	7	Sep 2017	Active	
5	7	Sep 2017	Active	
6	7	Feb 2017*	Active	
7	7	May 2018	Dropout	Aug 2019
8	7	May 2018	Dropout	Aug 2019
9	6	Sep 2018	Active	
10	7	Sep 2018	Active	
11	6	Sep 2018	Active	
12	6	Sep 2018	Dropout	Jan 2020
13	6	Sep 2018	Active	
14	7	Sep 2019	Active	

*Student 3 and Student 6 began violin training in July and February 2017, respectively, whereas formal data collection

began in September 2017. Therefore, their initial learning periods contain missing observations.

performance preparation periods were identified based on patterns observed in the collected data.

In addition to performance preparation, several external factors may have caused discrepancies between textbook progression and students' underlying learning levels. The first factor was external expectations, referring to situations in which parents, teachers, or other circumstances required students to advance to a specific textbook level beyond their current demonstrated ability. The second factor was lesson interruption due to vacations. Since lessons were conducted weekly, an absence of more than one week could affect learning continuity, while interruptions exceeding two weeks were expected to result in temporary declines in performance level.

In this study, students' underlying learning states were assumed to be influenced by textbook progression, prior knowledge (for age-7 beginners), performance preparation periods, external expectations, and lesson interruptions caused by vacations. Since students' true learning states cannot be directly observed, they were inferred from observable indicators, including textbook progression and learning-related issues. Therefore, a Dynamic Linear Model (DLM) based on the Kalman Filter framework was adopted as the time-series modeling approach to estimate students' latent learning trajectories.